T. Tony Cai, Weidong Liu, and Harrison H. Zhou. Estimating sparse precision matrix: Optimal rates of convergence and adaptive estimation. *The Annals of Statistics*, 44(2):455–488, 2016.

Jiahua Chen and Zehua Chen. Extended Bayesian information criteria for model selection with large model spaces. *Biometrika*, 95(3):759–771, 2008.

Francis X. Diebold and Kamil Yılmaz. On the network topology of variance decompositions: Measuring the connectedness of financial firms. *Journal of Econometrics*, 182(1):119–134, 2014.

Kristin J. Forbes and Roberto Rigobon. No contagion, only interdependence: Measuring stock market comovements. *The Journal of Finance*, 57(5):2223–2261, 2002.

Mario Forni, Marc Hallin, Marco Lippi, and Lucrezia Reichlin. The generalized dynamic-factor model: Identification and estimation. *Review of Economics and Statistics*, 82(4):540–554, 2000.

Marc Hallin and Roman Liška. Determining the number of factors in the general dynamic factor model. *Journal of the American Statistical Association*, 102(478):603–617, 2007.

Gary Koop, M. Hashem Pesaran, and Simon M. Potter. Impulse response analysis in nonlinear multivariate models. *Journal of Econometrics*, 74(1):119–147, 1996.

Dom Owens, Haeran Cho, and Matteo Barigozzi. fnets: An R package for network estimation and forecasting via factor-adjusted VAR modelling, 2024. R package version 0.1.6, CRAN.

Jie Peng, Pei Wang, Nengfeng Zhou, and Ji Zhu. Partial correlation estimation by joint sparse regression models. *Journal of the American Statistical Association*, 104(486):735–746, 2009.

H. Hashem Pesaran and Yongcheol Shin. Generalized impulse response analysis in linear multivariate models. *Economics Letters*, 58(1):17–29, 1998.

Hélène Rey. Dilemma not trilemma: The global financial cycle and monetary policy independence. In *Proceedings of the Economic Policy Symposium, Jackson Hole*, pages 285–333. Federal Reserve Bank of Kansas City, 2013.

Robert Tibshirani. Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society: Series B*, 58(1):267–288, 1996.

Hui Zou. The adaptive lasso and its oracle properties. *Journal of the American Statistical Association*, 101(476):1418–1429, 2006.

9 Data and Code Availability

The data (weekly ETF returns) and the full code are available at `https://github.com/ezellabra/ML4E-code`. The pipeline is written in Python; NETS and FNETS are estimated in R (packages `nets` 0.9.1 and `fnets` 0.1.6) through a command-line bridge. All parameters are in `config.yaml` and the whole analysis is reproduced with `make all` (or `python run_all.py`).

A Robustness

Table 3 reruns every method under alternative choices. The levels of Diebold–Yilmaz and lasso connectedness hardly move: between 79.7% and 81.2% and between 78.6% and 80.5% across all scenarios. NETS is the most sensitive to the window length (57.8% with 78-week windows, 63.4%

Table 3. Robustness: mean total connectedness (%) by method, share of DY connectedness lost once the common factor is removed (FNETS), and FNETS common-variance share.

Scenario	Windows	DY	Lasso VAR	NETS	FNETS (idio.)	Lost w/o factor (%)	Common var. (%)
Baseline (w=104, H=4)	179	81.0	80.0	61.5	47.2	41	58
H=2	179	80.9	80.0	61.5	47.2	41	58
H=8	179	81.0	80.1	61.5	47.2	41	58
w=78	185	81.2	79.6	57.8	50.5	38	58
w=130	172	81.1	80.5	63.4	46.2	43	58
Excluding SPY	179	80.0	78.9	59.9	45.9	42	58
Excluding windows touching 2020	141	79.7	78.6	58.8	47.2	41	55

Table 4. Static connectedness: one estimate on the whole period (full sample) and on the subsamples before and after 2020. These are single long-sample estimates, not averages over rolling windows, so they differ from Tables 2 and 3.

Sample	Method	Total conn. (%)	LR p.corr. dens. (%)
Full sample	DY	81.9	100.0
Full sample	Lasso VAR	82.1	100.0
Full sample	NETS	73.5	60.1
Full sample	FNETS (idio.)	34.2	0.0
2011-2019	DY	83.0	100.0
2011-2019	Lasso VAR	82.8	100.0
2011-2019	NETS	69.6	60.1
2011-2019	FNETS (idio.)	34.1	2.0
2021-2026	DY	73.5	100.0
2021-2026	Lasso VAR	73.7	100.0
2021-2026	NETS	56.0	52.3
2021-2026	FNETS (idio.)	39.6	13.7

with 130-week windows), as shorter windows leave the penalized estimator less information and make it keep fewer links. FNETS idiosyncratic connectedness ranges from 45.9% to 50.5%. The horizon matters very little: Diebold–Yilmaz changes by about 0.1 points between $H = 2$ and $H = 8$, and the sparse methods not at all, because their estimated lag coefficients are almost all zero and their variance decompositions are essentially driven by the contemporaneous covariance. Excluding the S&P 500 lowers Diebold–Yilmaz connectedness by one point, and removing every window that contains a week of 2020 (141 windows left) lowers Diebold–Yilmaz, lasso and NETS connectedness by one to three points and leaves FNETS unchanged, without changing the ranking of the methods.

The main quantities are stable across scenarios. The share of Diebold–Yilmaz connectedness lost once the common factor is removed lies between 38% (78-week windows) and 43% (130-week windows), against 41% in the baseline; the variance share of the common factor is 55–58%. The ordering of the comparison criteria is also unchanged: the correlation of total connectedness with Diebold–Yilmaz is at least 0.96 for the lasso and between 0.84 and 0.90 for NETS, and in every scenario the dense estimators are the most stable from one window to the next, followed by NETS and then FNETS.

The subsample estimates in Table 4 show a lower level of connectedness after 2020 than before: Diebold–Yilmaz connectedness falls from 83.0% in 2011–2019 to 73.5% in 2021–2026, and NETS from 69.6% to 56.0%. Idiosyncratic FNETS connectedness rises slightly (from 34.1% to 39.6%), while the variance share of the common factor falls from 58% to 47%. The decline after 2020 is therefore consistent with a weaker common component rather than with fewer idiosyncratic links. Figure 6 shows that, outside the windows that contain 2020, the time pattern of each method is the same as in the baseline.

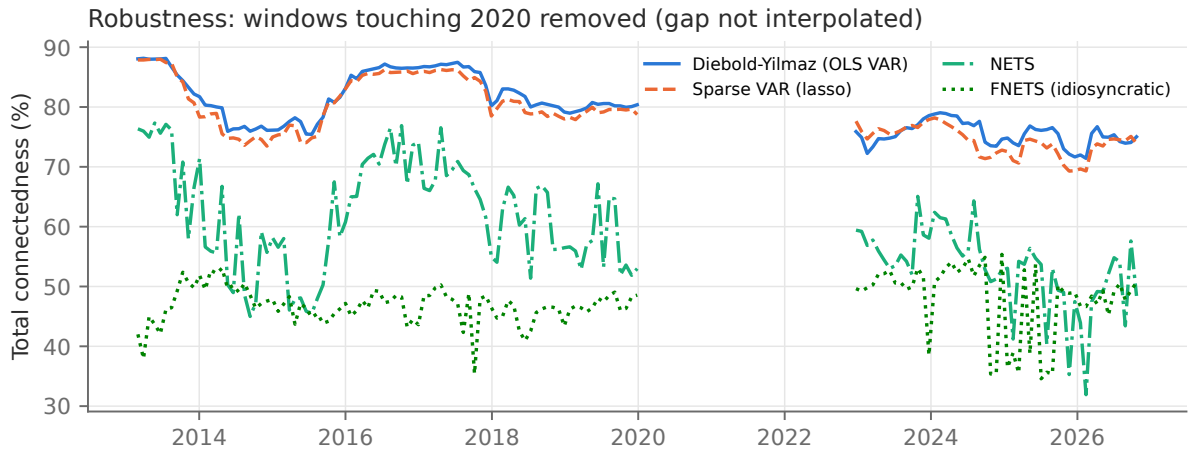


Fig. 6. Total connectedness excluding every window that contains a week of 2020.